INCH-POUND
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DETAIL SPECIFICATION

PRINTED WIRING ASSEMBLY, COMPUTER

This specification is approved for use by U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a Computer Printed Wiring Assembly, referred to herein as the PWA. This is one of six printed wiring assemblies that comprise the Simplified Test Equipment/Internal Combustion Engine-Reprogrammable (STE/ICE-R) Vehicle Test Meter (VTM) (see Drawing 12259265).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5998

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines (see 4.2.1).

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

DEPARTMENT OF DEFENSE

12259256 - Printed Wiring Assembly, Computer.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/IPC J-STD-001 - Requirements for Soldered Electrical and Electronic Assemblies.

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(Application for copies should be addressed to the American National Standard Institute, 11 West 42nd Street, New York, NY 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design, materials, and manufacturing processes. Design, materials, and manufacturing process selection shall be as specified herein and in applicable referenced specifications, standards, and drawings. Materials shall be uniform and free from imperfections or defects which affect their performance and serviceability. All metallic parts shall be made from corrosion resistant metals or treated with corrosion-resistant materials. Asbestos and Cadmium materials shall not be used in any form in any part of the PWA. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcuries per gram or activity per item equals or exceeds 0.01 microcuries.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Operating requirements. Each PWA shall provide the following functional, operational, and performance capabilities.

3.3.1 Input power. The PWA shall not require more than the current specified in table I.

TABLE I. Input power.

Source	Voltage	Current
P1-5, 6	+5 Vdc 1/	285 mA 2/

1/ Vdc = Volts direct current
2/ mA = milliampere

3.3.2 Complimentary metal oxide semiconductor (CMOS) logic levels. Unless otherwise specified (see 6.2), CMOS logic levels accepted by and output from the PWA shall be as follows:

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Input High (Logic Level 1) = 3.5 Vdc minimum (min.)
Input Low (Logic Level 0) = 1.5 Vdc maximum (max.)
Output High (Logic Level 1) = 4.5 Vdc min.
Output Low (Logic Level 0) = 0.5 Vdc max.

3.3.3 Input isolation. (NOTE: A "/" following a capitalized signal name denotes logic negation.)

3.3.3.1 EF1/ - EF4/, INTERRUPT/, and CLEAR/ isolation. Logic levels applied to any one of the P1-50, P1-52, P1-53, P1-54, P1-56, and P1-62, shall have no effect on nor be affected by logic levels applied at any of the other pins specified in this paragraph.

3.3.3.2 External input isolation. Logic levels applied to any one of P1-76, P1-77, P1-79, P1-81, P1-69, P1-71, P1-73, and P1-75 shall have no effect on nor be affected by logic levels applied at any of the other pins specified in this paragraph.

3.3.4 Reset.

3.3.4.1 Power-up reset. Signals at pins P1-68, U1-3, U13-1, U13-13, U22-1, U24-4, U24-10, and U25-4 shall conform to the waveforms shown in figure 1 as +5.0 Vdc is applied to P1-5 and P1-6.

3.3.4.2 CLEAR/. Applying a logic level 0 to P1-62 shall bring P1-68 to logic level 0 in less than 62 nanoseconds (ns).

3.3.4.3 States during power-up reset or CLEAR/. While P1-62 is held at logic level 0, the following pins shall be at the logic level designated in table II.

TABLE II. Logic levels during reset or CLEAR/.

Pin	Logic Level
P1-29	0
P1-30	0
P1-31	0
P1-32	0
P1-33	0
P1-34	0
P1-35	0
P1-37	0
P1-38	0
P1-72	1
U1-4, 17-20	0
U9-20	1

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TABLE II. Logic levels during reset or CLEAR/ - Continued.

Pin	Logic Level
U11 Pins	
5-7, 9-11, 12	0
U12-1	0
U13 Pins	
3, 5, 9, 11, 12	0
U14 Pins	
1, 7, 10	0
U22 Pins	
2, 5, 7, 9, 10, 12	0
U23 Pins	
13, 14, 15	0
U24 Pins 3, 11	0
U25-3	0
U27-15	1
P1-46	1
P1-64	1
P1-65	1
P1-74	1
P1-78	1
U1-2	1
U2 Pins	
1, 26	1
U3 Pins	
1, 26	1
U4 Pins	
1, 26	1
U5 Pins	
1, 26	1
U6 Pins	
1, 26	1
U7 Pins	
1, 26	1
U13-6	1
U17-1	1
U21-1	1
U24 Pins	
5, 9	1
U25-5	1
U26 Pins	
7, 9, 10,	
12-15	1
U27-13	1

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3.3.4.4 Power-up/down memory protection. Whenever the voltage at P1 pins 5 and 6 goes below 4.5 V, U33-5 shall maintain a voltage of no more than 0.4 V until the voltage at P1 pins 5 and 6 is less than 3.0 V. The voltage at U33-5 shall track the voltage at P1 pins 5 and 6. If the voltage at P1 pins 5 and 6 increases to a voltage greater than 4.6 V from a voltage less than 4.5 V, U33-5 shall maintain an output voltage of not greater than 0.4 V for a period of not less than 208 milliseconds (ms) and return to the voltage present at P1 pins 5 and 6 within 312 ms.

3.3.5 Clock frequencies.

3.3.5.1 Clock oscillator. The frequency of the signal at Y1-8, U12-3, and U15-1 shall be 4.000 megahertz (MHz) ± 4.00 kilohertz (kHz).

3.3.5.2 2 MHz clock. The frequency of the signal at P1-83 and U12-2 shall be 2.000 MHz ± 400 Hz.

3.3.5.3 400 kHz clock. The frequency of the signal at P1-67 and U14-2 shall be 400.0 kHz ± 400 Hz.

3.3.5.4 200 kHz clock. The frequency of the signal at P1-70 shall be 200.0 kHz ± 200 Hz.

3.3.6 Microprocessor outputs and inputs.

3.3.6.1 High order byte address latch. When TPA (U1-34) goes from logic level 1 to logic level 0, the logic levels present at U1-25 through U1-32 shall be latched into U10 and shall be present at pins, 2, 5, 6, 9, 12, 15, 16 and 19, respectively, after the falling edge of TPA.

3.3.6.2 Microprocessor inputs. A logic level 0 applied at any microprocessor input (P1-49 through P1-54 and P1-56) shall cause the corresponding pins of U1 (38, 36, 37, 24, 21, 23, and 22, respectively) to be at logic level 0. The applied logic level shall be capable of sinking not less than 175 microamperes (µA).

3.3.7 Board logic and functions.

3.3.7.1 I/O port decoder. U11 shall have outputs shown in table III at P1-29 through P1-35 for the corresponding inputs at U11 pins 2, 3, and 14 between the time that TPA (U1-34) goes from logic level 1 to logic level 0 and the time that TPB (U1-33) goes from logic level 0 to logic level 1.

3.3.7.2 OEN-Q. The OEN-Q signal shall conform to the waveform in figure 2 when the Q output (U1-4) is at logic level 1.

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3.3.7.3 OEN-Q/. The OEN-Q/ signal shall conform to the waveform in figure 3 when the Q output (U1-4) is at logic level 0.

TABLE III. I/O port decoder logic.

U11 Pin			I/O decoder output P1-						
14	3	2	29	30	31	32	33	34	35
0	0	0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0	0	0
0	1	0	0	1	0	0	0	0	0
0	1	1	0	0	1	0	0	0	0
1	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	1

3.3.7.4 Data BUS transceiver logic. The data BUS transceiver direction pin (U30-8) shall be at logic level 0 when MRD/ (U19-14) and any of the U1 pins 17, 18, or 19 are at logic level 1. Otherwise, U30-8 shall be at logic level 1.

3.3.7.5 Data BUS inputs. The logic levels present at P1-21 through P1-28 shall appear at U21 pins 9, 8, 7, 6, 5, 4, 3, and 2, respectively, when U30-8 is at logic level 0. Otherwise, logic levels present at U21 pins 9, 8, 7, 6, 5, 4, 3, and 2 shall appear at P1-21 through P1-28.

3.3.7.6 External input logic. The external input output enable pin (U30-11) shall be at logic level 0 when IOB/3 (U11-7) and MRD/ (U19-14) are at logic level 1. Otherwise, U30-11 shall be at logic level 1.

3.3.7.7 External inputs. Logic levels applied at P1-76, P1-77, P1-79, P1-81, P1-69, P1-71, P1-73, and P1-75 shall appear at U17 pins 12, 9, 5, 6, 16, 15, 19, and 2 when U30-11 is at logic level 0. U17 pins 12, 9, 5, 6, 16, 15, 19, and 2 shall not drive the data BUS otherwise.

3.3.7.8 Extended I/O latches and decoder. Logic levels present at U1 pins 15 through 11 shall be latched into U22 and shall be present at U22 pins 2, 5, 7, 10, and 12 respectively, at the rising edge of OEN-Q (U28-12) when IOF/7 (U11-9) is at logic level 1. The logic levels present at U23 pins 13, 14, and 15 shall be determined by the logic levels at U22 pins, 2, 5, 7, and 10, according to table IV.

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TABLE IV. Extended I/O decoder outputs.

U22 Pin					U23 Pin			Port	
12	10	7	5	2	13	14	15	address	Function
X	0	X	X	X	0	0	0	---	---
0	1	0	0	0	0	0	1	08 Hex	2 MHz CPU
1	1	0	0	0	0	0	1	18 Hex	4 MHz CPU
X	1	0	0	1	0	1	0	09 Hex	CNTR
X	1	0	1	0	1	0	0	0A Hex	MSEL
X	1	X	1	1	0	0	0	---	---
X	1	1	X	X	0	0	0	---	---

(X denotes a don't care condition)

3.3.7.9 Extended I/O data clocking. Logic levels present at U1 pins 8 through 15 shall be clocked into a previously selected extended I/O port (certain data BUS bits may not be connected) at the rising edge of OEN-Q/ U28-6) when IOF/7 (U11/9) is at logic level 1. Table V shows the active port clock as a function of the selected extended I/O port.

TABLE V. Extended I/O port clocks.

Selected port address	Function	Port clock
08 Hex	2 MHz CPU	U28-8
18 Hex	4 MHz CPU	U28-8
09 Hex	CNTR	U29-12
0A Hex	MSEL	U29-6

3.3.7.10 Microprocessor clock frequency select. The clock input to the microprocessor shall be able to change between 2 MHz and 4 MHz frequencies according to the waveforms in figure 4 when selected by the requirement of paragraph 3.3.7.8. Paragraph 3.3.7.9 shall be used to clock the frequency selection of paragraph 3.3.7.8 using "don't care" conditions for data.

3.3.7.11 Memory bank select. The memory bank select line, MSEL (U25-5), when selected by the requirement of paragraph 3.3.7.8, shall have a logic level equivalent to the logic level at U1-15 when clocked by the requirement of paragraph 3.3.7.9.

3.3.7.12 Bootstrap EPROM page select. The control lines, CNTR1 (U24-5) and CNTR2 (U24-9), when selected by the requirement of paragraph 3.3.7.8, shall have logic levels equivalent to the logic levels at U1 pins 15 and 14, respectively, when clocked by the requirement of paragraph 3.3.7.9.

3.3.7.13 Memory chip select. The memory chip select line U26 pins 13, 7, 9, 10, 12, 14, and 15 and U27 pins 13 and 15 shall have the logic levels shown in tables VI and VII for the output from U10 pins 15, 16, and 19, U33-5, and U25-5.

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TABLE VI. Memory chip select logic levels.

U25-5	U33-5	U10 pin 19	U10 pin 16	U10 pin 15	U26-15 U2-20	U26-14 U3-20	U26-13 U8-20	U26-12 U4-20	U26-10 U5-20	U26-9 U6-20	U26-7 U7-20
X	0	X	X	X	1	1	1	1	1	1	1
1	X	X	X	X	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1	1	1
0	1	0	0	1	1	0	1	1	1	1	1
0	1	0	1	0	1	1	0	1	1	1	1
0	1	0	1	1	1	1	1	0	1	1	1
0	1	1	0	0	1	1	1	1	1	1	1
0	1	1	0	1	1	1	1	1	0	1	1
0	1	1	1	0	1	1	1	1	1	0	1
0	1	1	1	1	1	1	1	1	1	1	0

TABLE VII. Memory chip select logic levels.

U25-5	U10 pin 15	U10 pin 16	U10 pin 19	U27-15 U9-20	U27-13 U8-20
0	X	X	X	1	1
1	0	0	0	0	1
1	0	0	1	1	1
1	0	1	0	1	0
1	0	1	1	1	1
1	1	X	X	1	1

(X denotes a don’t care conditions)

3.3.8 Memory.

3.3.8.1 Bootstrap EPROM read. U9 pins 11 through 13, and 15 through 19 shall drive the data BUS with logic levels defined by the logic levels present at U9 pins 1 through 10, 21, and 24 through 27, (see 6.3) when the control signals (MRD/ - U19-14, and EPS/ - U27-15) conform to the waveforms in figure 5. Logic levels at U9 pins 26 and 27 are determined by logic levels at CNTR1 (U24-5) and CNTR2 (U24-9), respectively. See table VIII for the address range for each control line setting.

3.3.8.2 SRAM write. U8 shall store the logic levels present at U1 pins 8 through 15 in a location defined by the logic levels present at U8 pins 2 through 10, 21, and 23 through 25 when the control signals (MWR/ - U20-17 and RS/ - U32-6) conform to the waveform in figure 6.

TABLE VIII. EPROM addressing.

CNTR1 U24-5	CNTR2 U24-9	U9 address range (Hex)
0	0	0000 - 1FFF
0	1	2000 - 3FFF
1	0	4000 - 5FFF
1	1	6000 - 7FFF

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3.3.8.3 SRAM read. U8 pins 11 through 13, and 15 through 19 shall drive the data BUS with logic levels defined by both the logic levels present at U8 pins 2 through 10, 21, 23 through 25, and data written by the requirement of paragraph 3.3.8.2, when the control signals (MRD/ - U19-14 and RS/ -32-6) conform to the waveforms in figure 7.

3.3.8.4 EEPROM byte write. EPROM's, U2 through U7, shall be each capable of storing the logic levels present at U1 pins 8 through 15 in a location defined by the logic levels present at each EEPROM's address inputs (pins 2 through 10, 21, and 23 through 25) when MWR/ (U20-17) and a chip select from table IX conform to the waveforms in figure 8.

TABLE IX. EEPROM chip select signals.

Signal name	Chip select line	Selected device
CS1/	U26-15	U2
CS2/	U26-14	U3
CS3/	U26-12	U4
CS4/	U26-10	U5
CS5/	U26-9	U6
CS6/	U26-7	U7

3.3.8.5 EEPROM read. EEPROM's, U2 through U7, shall each be able to drive the data BUS with logic levels defined by both the logic levels present at each EEPROM's address inputs (pins 2 through 10, 21, and 23 through 25), and data written by the requirement of paragraph 3.3.8.4 when MRD/ (U19-14) and a chip select from table IX conform to the waveforms in figure 9.

3.3.8.6 EEPROM page write. EEPROM's, U2 through U7, shall each be capable of storing up to 16 bytes (the logic levels present at U1 pins 8 through 15) in locations defined by the logic levels present at each EEPROM's address inputs (pins 2 through 10, 21, and 23 through 25; logic levels at pins 2 through 10, 21, and 23 through 25 must remain the same) when MWR/ (U20-17) and a chip select from table IX conform to the waveforms in figure 10.

3.3.9 ATE Test. The PWA shall pass TACOM approved ATE test installed in TACOM approved ATE or equivalent.

3.3.10 Soldering. Soldering shall meet or exceed the requirements of ANSI/IPC J-STD-001.

3.4 Interface requirements.

3.4.1 Overall envelope. Overall envelope of the PWA shall be in accordance with drawing 12259256.

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3.5 Ownership and support requirements. The PWA shall meet the following logistics requirements.

3.5.1 Interchangeability. Interchangeability tolerances should permit parts, subassemblies and assemblies to be used in their parent assemblies without regard to the source of supply or manufacturer. Parts, subassemblies and assemblies having the full range of dimensions and characteristics permitted by the specification governing the part, subassembly or assembly should be usable as replacement items without selection and without departure from the specified performance guidelines of the parent items.

3.5.2 Identification marking. Unless otherwise specified (see 6.2), identification marking shall be permanent and legible and shall include as a minimum the following:

a. Nomenclature “PRINTED WIRING ASSEMBLY, COMPUTER”.
b. Military part number.
c. Manufacturer’s CAGE code and name.
d. Manufacturer’s serial number.
e. Contract number.
f. Symbol “US”.

3.6 Operating environment requirements. The PWA shall operate under the following environmental condition, without degradation in performance:

3.6.1 Temperature.

3.6.1.1 High temperature (operating). The PWA shall be capable of operation without damage or malfunction at a high temperature of 125°F (52°C) (without solar radiation).

3.6.1.2 Low temperature (operating). The PWA shall be capable of operation without damage or malfunction at a low temperature of 20°F (-7°C). For ambient temperature below 20°F (-7°C), the test set shall be operated in a shelter.

3.6.1.3 High temperature (nonoperating). The PWA shall be capable of being stored and transported in its case without damage at an ambient temperature of plus 160°F (71°C).

3.6.1.4 Low temperature (nonoperating). The PWA shall be capable of being stored and transported in its case without damage in ambient temperatures of minus 60°F (-51°C).

3.6.2 Fungus. All nonmetallic materials not inherently fungistatic shall function in a fungus environment without adversely affecting the operation/performance of PWA.

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3.6.3 Shock. The PWA shall evidence no degradation of function and meet the requirements of 3.3 when subjected to the shock test.

3.6.4 Vibration. The PWA shall evidence no degradation of function and meet the requirements of 3.3 when subjected to the vibration test.

3.6.5 Burn-in. Unless otherwise specified (see 6.2), the PWA shall be subjected to and pass a "power on" burn-in prior to the inspections listed in 4.1.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspection (CI) (see 4.1.2).

4.1.1 First article inspection. Unless otherwise specified (see 6.2), the Government shall select one PWA produced under the production contract for first article inspection. First article sample shall include all the verifications of table X.

4.1.2 CI 100% inspection. CI shall be conducted on all items (100% inspection) and shall include those examinations and tests from table X, as specified in the contract.

TABLE X. Verification methods.

Title	Requirement	Verifica-tion
Design, materials, & manufacturing processes	3.2	4.2
Operating requirements:	3.3	4.2.3
Input power	3.3.1	4.2.3.1
CMOS Logic levels	3.3.2	4.2.3.2
EF1/ -EF4/, INTERRUPT/, and CLEAR/ isolation	3.3.3.1	4.2.3.3.1
External input isolation	3.3.3.2	4.2.3.3.2
Power-up reset	3.3.4.1	4.2.3.4.1
Clear/	3.3.4.2	4.2.3.4.2
States during power-up reset or CLEAR/	3.3.4.3	4.2.3.4.3
Power-up/down memory protection	3.3.4.4	4.2.3.4.4
Clock oscillator	3.3.5.1	4.2.3.5

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TABLE X. Verification methods - Continued.

Title	Requirement	Verification
2 MHz clock	3.3.5.2	4.2.3.5
400 kHz clock	3.3.5.3	4.2.3.5
200 kHz clock	3.3.5.4	4.2.3.5
High order byte address latch	3.3.6.1	4.2.3.6.1
Microprocessor inputs	3.3.6.2	4.2.3.6.2
I/O port decoder	3.3.7.1	4.2.3.7.1
OEN-Q	3.3.7.2	4.2.3.7.2
OEN-Q/	3.3.7.3	4.2.3.7.3
Data BUS transceiver logic	3.3.7.4	4.2.3.7.4
Data BUS inputs	3.3.7.5	4.2.3.7.5
External input logic	3.3.7.6	4.2.3.7.6
External inputs	3.3.7.7	4.2.3.7.7
Extended I/O latches and decoder	3.3.7.8	4.2.3.7.8
Extended I/O data clocking	3.3.7.9	4.2.3.7.9
Microprocessor clock frequency select	3.3.7.10	4.2.3.7.10
Memory bank select	3.3.7.11	4.2.3.7.11
Bootstrap EPROM page select	3.3.7.12	4.2.3.7.12
Memory chip select	3.3.7.13	4.2.3.7.13
Bootstrap EPROM read	3.3.8.1	4.2.3.8.1
SRAM write	3.3.8.2	4.2.3.8.2
SRAM read	3.3.8.3	4.2.3.8.3
EEPROM byte write	3.3.8.4	4.2.3.8.4
EEPROM read	3.3.8.5	4.2.3.8.5
EEPROM page write	3.3.8.6	4.2.3.8.6
ATE Test 1/	3.3.9	4.2.3.9
Interface requirements	3.4	4.2.4
Overall enveloped & interfaces	3.4.1	4.2.4.1
Ownership & support requirements	3.5	4.2.5
Interchangeability	3.5.1	4.2.5.1
Identification marking	3.5.2	4.2.5.2

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TABLE X. Verification methods - Continued.

Title	Requirement	Verification
Operating environment requirements:	3.6	4.2.6
Temperature	3.6.1	4.2.6.1
Fungus	3.6.2	4.2.6.2
Shock	3.6.3	4.2.6.3
Vibration	3.6.4	4.2.6.4

1/ Not required unless specified (see 6.2).

4.2 Verification methods. Acceptable verification methods included in this section are visual inspection, and measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.2.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures to verify performance. See the contract for alternatives that replace verification methods required by this specification.

4.2.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed under the following standard (room) ambient conditions:

a. Temperature: 77 ± 18°F (25 ± 8°C).
b. Relative humidity: Uncontrolled room ambient.
c. Atmospheric pressure: Site pressure.

4.2.3 Operating requirements verification.

4.2.3.1 Input power. To determine conformance to 3.3.1, the input current shall be measured with an ammeter.

4.2.3.2 CMOS logic levels. To determine conformance to 3.3.2, the input and output CMOS logic levels shall be observed and verified with an oscilloscope.

4.2.3.3 Input isolation.

4.2.3.3.1 EF1/ - EF4/, INTERRUPT/, and CLEAR/ isolation. To determine conformance to 3.3.3.1, the following procedures shall be performed:

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a. Connect a 10k (where k = 1000) ± 10% ohm resistor to each of the board pins P1-50, P1-52, P1-53, P1-54, P1-56, and P1-62.
b. Apply 5.00 V to the unterminated end (the test input) of the 10 kilohms (kohm) resistor connected to P1-50. Apply 0.00 V to the other seven test inputs.
c. Verify that the voltage at P1-50 is not less than 4.5 V and the voltages at the other seven test inputs are not greater and 0.5 V.
d. Repeat parts (b) and (c) for P1-52, P1-53, P1-54, P1-56, and P1-62.

4.2.3.3.2 External input isolation. To determine conformance to 3.3.3.2, the following procedures shall be performed:

a. Connect a 10k ± 10% ohm resistor to each of the board pins P1-76, P1-77, P1-79, P1-81, P1-69, P1-71, P1-73, and P1-75.
b. Apply 5.00 V to the unterminated end (the test input) of the 10 kohm resistor connected to P1-76. Apply 0.00 V to the other seven test inputs.
c. Verify that the voltage at P1-76 is not less than 4.5 V and the voltages at the other seven test inputs are not greater than 0.5 V.
d. Repeat parts (b) and (c) for P1-77, P1-79, P1-81, P1-69, P1-71, P1-73, and P1-75.

4.2.3.4 Reset.

4.2.3.4.1 Power-up reset. To determine conformance to 3.3.4.1, +5.00 Vdc shall be applied to P1-5 and P1-6 and the voltage shall be observed at P1-68, U30-6, U1-3, U13 pins 1, 13, U22-1, U24 pins 4, 10, and U25-4 for compliance with figure 1.

4.2.3.4.2 CLEAR/. To determine conformance to 3.3.4.2, a logic level 0 shall be applied to P1-62 and the logic level shall be observed at P1-68.

4.2.3.4.3 States during power-up reset or CLEAR/. To determine conformance to 3.3.4.3, a logic level 0 shall be maintained at P1-68 and the logic levels shall be observed at the pins specified in table II.

4.2.3.4.4 Power-up/down memory protection. To determine conformance to 3.3.4.4, 5.0 Vdc shall be applied to P1 pins 5 and 6 for at least 1 second and the voltage at U33-5 shall be verified to be 5 Vdc. The voltage at P1 pins 5 and 6 shall be reduced to a voltage less than 4.5 Vdc. The voltage applied to P1 pins 5 and 6 shall be increased to a voltage between 4.6 Vdc and 5.0 Vdc. The voltage at U33-5 shall be verified to maintain a voltage not greater than 0.4 Vdc for at least 208 ms and return to the voltage applied to P1 pins 5 and 6 within 312 ms.

4.2.3.5 Clock frequencies. To determine conformance to 3.3.5.1 through 3.3.5.4, the frequency of the signal as required shall be observed with an oscilloscope.

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4.2.3.6 Microprocessor outputs and inputs.

4.2.3.6.1 High order byte address latch. To determine conformance to 3.3.6.1, a logic analyzer shall be used to verify that the logic levels present at U1-25 through U1-32 are present at U10 pins 2, 5, 6, 9, 12, 15, 16, and 19 after the falling edge of TPA (U1-34).

4.2.3.6.2 Microprocessor inputs. To determine conformance to 3.3.6.2, a logic level 0 shall be applied at P1-49 through P1-54 and P1-56. A logic level 0 shall be verified to be present at U1 pins 38, 36, 37, 24, 21, 23, and 22, respectively. An ammeter shall be used to verify that the current at P1-49 through P1-54 and P1-56 is not greater than 175 microamperes.

4.2.3.7 Board logic and functions.

4.2.3.7.1 I/O port decoder. To determine conformance to 3.3.7.1, the logic levels at U11 pins 5 through 7 and 9 through 12 shall be verified to correspond to the logic levels present at U1-19, U-18, and U1-17 given in table V between the falling edge of TPA (U1-34) and the rising edge of TPB (U1-33).

4.2.3.7.2 OEN-Q. To determine conformance to 3.3.7.2, a logic analyzer shall be used to verify that the signal at U28-12 conforms to the waveforms of figure 2 when the Q output (U1-4) is set to a logic level 1.

4.2.3.7.3 OEN-Q/. To determine conformance to 3.3.7.3, a logic analyzer shall be used to verify that the signal at U26-6 conforms to the waveforms of figure 3 when the Q output (U1-4) is set to a logic level 0.

4.2.3.7.4 Data BUS transceiver logic. To determine conformance to 3.3.7.4, a logic analyzer shall be used to verify that a logic level 0 is present at U30-8 when U19-14 and either U1-19, U1-16, or U1-17 are at logic level 1.

4.2.3.7.5 Data BUS inputs. To determine conformance to 3.3.7.5, a logic analyzer shall be used to verify that the logic levels applied at P1-21 through P1-28 are present at U21 pins 9, 8, 7, 6, 5, 4, 3, and 2 only when U30-8 is at logic level 0.

4.2.3.7.6 External input logic. To determine conformance to 3.3.7.6, a logic analyzer shall be used to verify that a logic level 0 is present at U30-11 when U11-7 and U19-14 are at logic level 1.

4.2.3.7.7 External inputs. To determine conformance to 3.3.7.7, a logic analyzer shall be used to verify that the logic levels present at P1-76, P1-77, P1-79, P1-81, P1-69, P1-71, P1-73, and P1-75 are present at U17 pins 12, 9, 5, 6, 16, 15, 19, and 2 only when U30-11 is at logic level 0.

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4.2.3.7.8 Extended I/O latches and decoder. To determine conformance to 3.3.7.8, a logic analyzer shall be used to verify that logic levels present at U1 pins 15 through 11 appear at U22 pins 2, 5, 7, 10, and 12 after the rising edge of OEN-Q (U28-12) when IOF/7 (U11-9) is at logic level 1. The logic levels at U23, pins 15 through 13, shall be verified to conform to the requirements of table IV.

4.2.3.7.9 Extended I/O data clocking. To determine conformance to 3.3.7.9, a logic analyzer shall be used to verify that the rising edge of the signal at the active port clock, as specified by table V, occurs after the rising edge of OEN-Q/ when IOF/7 is at logic level 1.

4.2.3.7.10 Microprocessor clock frequency select. To determine conformance to 3.3.7.10, a logic analyzer shall be used to verify that the signals at U13 pins 1, 3, 5, 6, 9, 11, and 12 conform to the waveforms of figure 4 when selected by the requirements of 3.3.7.8 and clocked by the requirements of 3.3.7.9.

4.2.3.7.11 Memory bank select. To determine conformance to 3.3.7.11, a logic analyzer shall be used to verify that the level at U25-5 is equivalent to the logic level at U1-15 when selected by the requirements of paragraph 3.3.7.8 and clocked by the requirements of paragraph 3.3.7.9.

4.2.3.7.12 Bootstrap EPROM page select. To determine conformance to 3.3.7.12, a logic analyzer shall be used to verify that the logic levels at U24-5 and U24-9 are equivalent to the logic levels at U1-15 and U1-14 when selected by the requirements of 3.3.7.8 and clocked by the requirements of 3.3.7.9.

4.2.3.7.13 Memory chip select. To determine conformance to 3.3.7.13, a logic analyzer shall be used to verify that the memory chip select outputs conform to the requirements of tables VI and VII.

4.2.3.8 Memory.

4.2.3.8.1 Bootstrap EPROM read. To determine conformance to 3.3.8.1, a logic analyzer shall be used to verify the logic levels on the data BUS (see 6.3) when valid address and control signals are supplied to U9 as shown in the waveforms of figure 5.

4.2.3.8.2 SRAM write. To determine conformance to 3.3.8.2, U8 shall be written to by supplying the control signals and valid data as shown in the waveforms of figure 6. The method of 4.2.3.8.3 shall be used for verification.

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4.2.3.8.3 SRAM read. To determine conformance to 3.3.8.3, a logic analyzer shall be used to verify that the logic levels on the data BUS conform to those written by the method of 4.2.3.8.2 when valid address and control signals are supplied to U8 as shown in the waveforms of figure 7.

4.2.3.8.4 EEPROM byte write. To determine conformance to 3.3.8.4, each EEPROM (U2 through U7) shall be written to by supplying the control signals and valid data as shown in the waveforms of figure 8. The method of 4.2.3.8.5 shall be used for verification.

4.2.3.8.5 EEPROM read. To determine conformance to 3.3.8.5, a logic analyzer shall be used to verify that the logic levels on the data BUS conform to those written by the method of 4.2.3.8.4 or 4.2.3.8.6 when valid address and control signals are supplied to each EEPROM (U2 through U7) as shown in the waveforms of figure 9.

4.2.3.8.6 EEPROM page write. To determine conformance to 3.3.8.6, each EEPROM (U2 through U7) shall be written to by supplying the control signals and valid data as shown in the waveforms of figure 10. The method of 4.2.3.8.5 shall be used for verification.

4.2.3.9 ATE test. To determine conformance to 3.3.9, install PWA into TACOM approved fixture (such as BTF-2500-92), on TACOM approved, ATE equipment (such as HP 3062) with TACOM approved software and instructions (such as BSW-25000920).

4.2.4 Interface requirements verification.

4.2.4.1 Overall envelope inspection. Insert the PWA into the present existing location of the test set, verify its envelope and interfacing conformance.

4.2.5 Ownership and support requirements verification.

4.2.5.1 Interchangeability inspection. To determine conformance to 3.5.1, the interchangeability between different parts, subassemblies, and assemblies shall be verified.

4.2.5.2 Identification marking inspection. Verify the presence of the required PWA identification marking.

4.2.6 Operating environment requirements verification performed for first article. The electrical performance tests specified in 4.2.3.1 through 4.2.3.8.6 shall be conducted prior to and at the completion of each environmental test. Verification of performance during the high temperature and low temperature environmental tests shall be demonstrated by the tests specified in 4.2.3.3 through 4.2.3.8.6.

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4.2.6.1 Temperature. To determine conformance to 3.6.1, the PWA shall be tested as follows.

4.2.6.1.1 High temperature (operating). To determine conformance to 3.6.1.1, the PWA shall be subjected to high temperature test in accordance with method 501.3, procedure II (hot category), of MIL-STD-810, or equivalent (see 4.2.1), and tested to 3.3.

4.2.6.1.2 Low temperature (operating). To determine conformance to 3.6.1.2, the PWA shall be subjected to low temperature test in accordance with method 502.3, procedure II, of MIL-STD-810, or equivalent (see 4.2.1), and tested to 3.3.

4.2.6.1.3 High temperature (nonoperating). To determine conformance to 3.6.1.3, the PWA shall be subjected to high temperature test in accordance with method 501.3, procedure I, of MIL-STD-810, or equivalent (see 4.2.1).

4.2.6.1.4 Low temperature (nonoperating). To determine conformance to 3.6.1.4, the PWA shall be subjected to low temperature test in accordance with method 502.3, procedure I, of MIL-STD-810, or equivalent (see 4.2.1).

4.2.6.2 Fungus. To determine conformance to 3.6.2, the PWA shall be subjected to the applicable fungus test in accordance with method 508.4 of MIL-STD-810 (28 day test), or equivalent (see 4.2.1).

4.2.6.3 Shock. To determine conformance to 3.6.3, the PWA shall be subjected to the shock test in accordance with method 516.4, Procedure I, 40g peak of MIL-STD-810, or equivalent (see 4.2.1), and tested to 3.3 after shock test.

4.2.6.4 Vibration. To determine conformance to 3.6.4, the PWA shall be subjected to a vibration test using a composite of figure 11 through 13. These figures represent the cargo environment at the cargo bed of a composite of tactical wheeled vehicles, the 5-ton M813 truck and the 12-ton M127 semitrailer. The data include differing vehicle loading conditions traversing over specially designed courses ranging from paved highway to off road conditions at various vehicle speeds. The spectrum is broad band random with peaks and notches at various discrete frequency bands. Break points are provided for establishing the spectrum shape. The test duration is 500 miles. The PWA shall be tested to 3.3 after the vibration test.

4.2.6.5 Burn-in. When required (see 6.2), the burn-in shall consist of three temperature cycles and a six gravity unit (g) root mean square (rms) random vibration as follows:

a. Temperature cycle. Each temperature cycle shall consist of 30 minutes at minus (-) 60 degrees Fahrenheit (°F) (-51 degrees Celsius (°C)) and 30 minutes at plus

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(+) 125°F (52°C). The rate of temperature change between temperature extremes shall be not less than 5°F (2.8°C) per minute. Power shall be applied to the PWA for not less than 30 minutes from 20°F (7°C) through the completion of each high-temperature soak.

b. Random vibration. Following all temperature cycling, the PWA shall be hard mounted to the vibration table so that the axis of vibration is perpendicular to the plane of the PWA. The vibration shall be at room ambient temperature, for a period of 10 minutes, as shown in figure 14.

NOTE: PWAs being procured for simultaneous installation in VTMs being procured do not have to be subjected to “burn-in” provided the VTMs are subjected to the 16-hour “power on” burn-in.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department or Defense Agency, or within the Military Department’s System Command. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The printed wiring assemblies of the STE/ICE-R vehicle test set determine primary fault isolation of a vehicle engine, fuel or electrical system. The PWA contains an 8-bit microprocessor with 16-bit memory addressing and has an interrupt input and four-event flag inputs. There are two banks of memory; a 23k by 8 ultraviolet Erasable Programmable PROMs which are removed for updated and reprogrammable memory via another VTM (EEPROM’s; U2-U7). An 8k by 8 Static Random Access Memory (SRAM; U8) is accessible from both banks. The microprocessor runs on either a 4 MHz or 2 MHz clock, which is software selectable. Logic levels on the external input lines can be gated onto the data BUS (pins 21 through 28; board and IC pins will be designated P1-XX and UZ-XX, where XX is the pin number and Z is the IC number). A bus transceiver isolates the microprocessor and memory from the rest of the data BUS. The PWA is military unique because it is designed specifically to be a component of the Simplified Test Equipment/Internal Combustion Engine - Reprogrammable

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(STE/ICE-R) Vehicle Test Meter (VTM), and to withstand environmental conditions which exceed commercial requirements.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. If first article inspection is required (see 3.1).
d. If CMOS logic levels should be other than as specified (see 3.3.2).
e. PWA drawing number, revision letter, date and PWA part number (see 3.2 and 3.5.2).
f. If marking is other than as specified (see 3.5.2).
g. If burn-in is not required (see 3.6.5).
h. Whether ATE test is required (see table X footnote).
i. Packaging requirements (see 5.1).

6.3 Program index. Drawing 12259329/IPI, Program index is being used to define the logic levels on the U9 pins.

6.4 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.5 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table X and their designated frequency based on a risk assessment for the procurement.

6.6 Historical data. Historically, ATPD 2145, dated 9 March 1988, was superseded by MIL-P-62737(AT).

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6.7 Subject term (key word) listing.
Automated Testing Equipment (ATE) test
Ignition inhibits
Ignition signal outputs
Simplified Test Equipment/Internal Combustion Engine - Reprogrammable (STE/ICE-R)
Vehicle Test Meter (VTM)

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

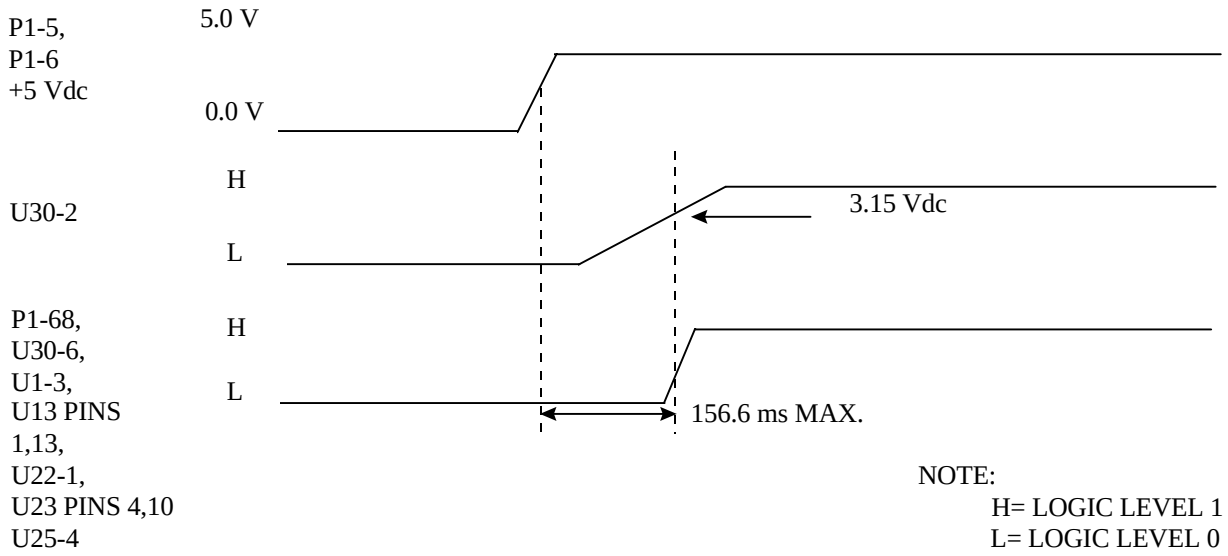


FIGURE 1. Power-up reset and CLEAR/ waveforms.

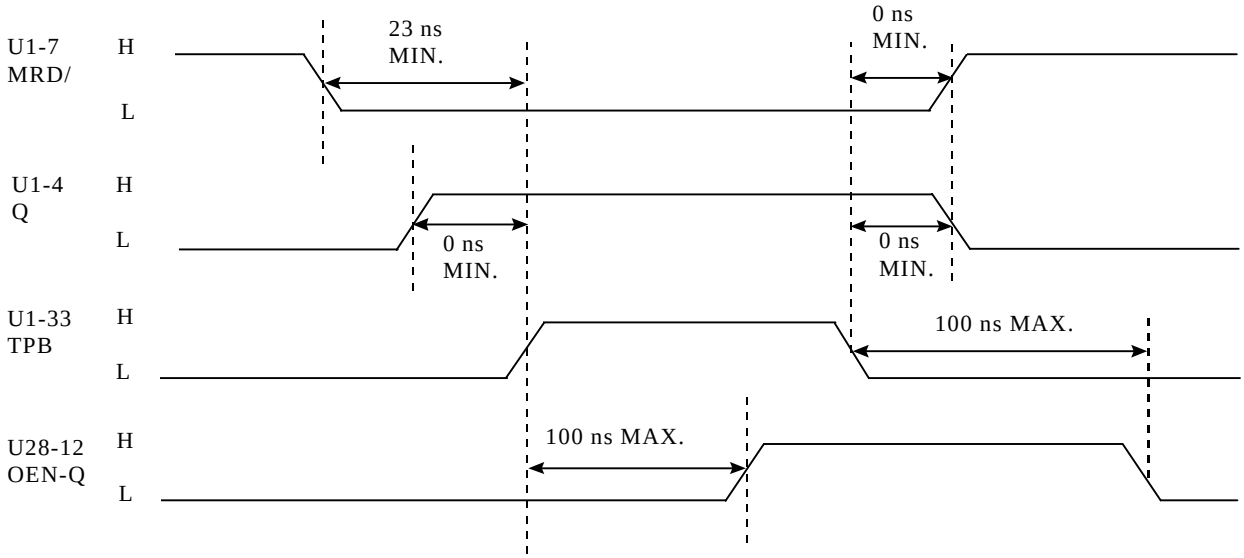
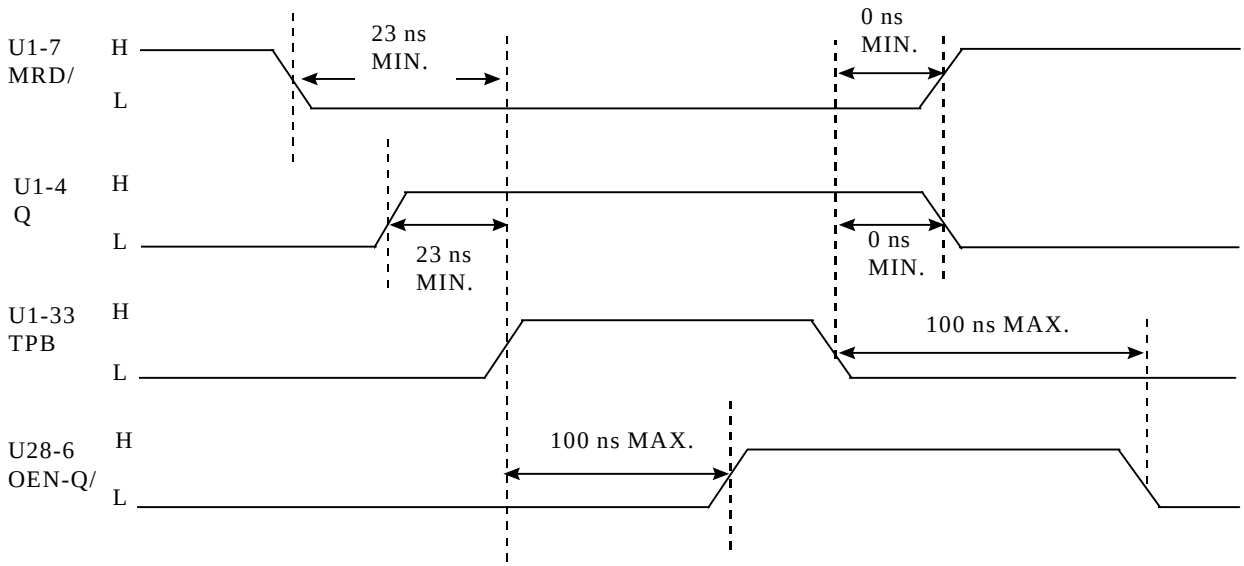


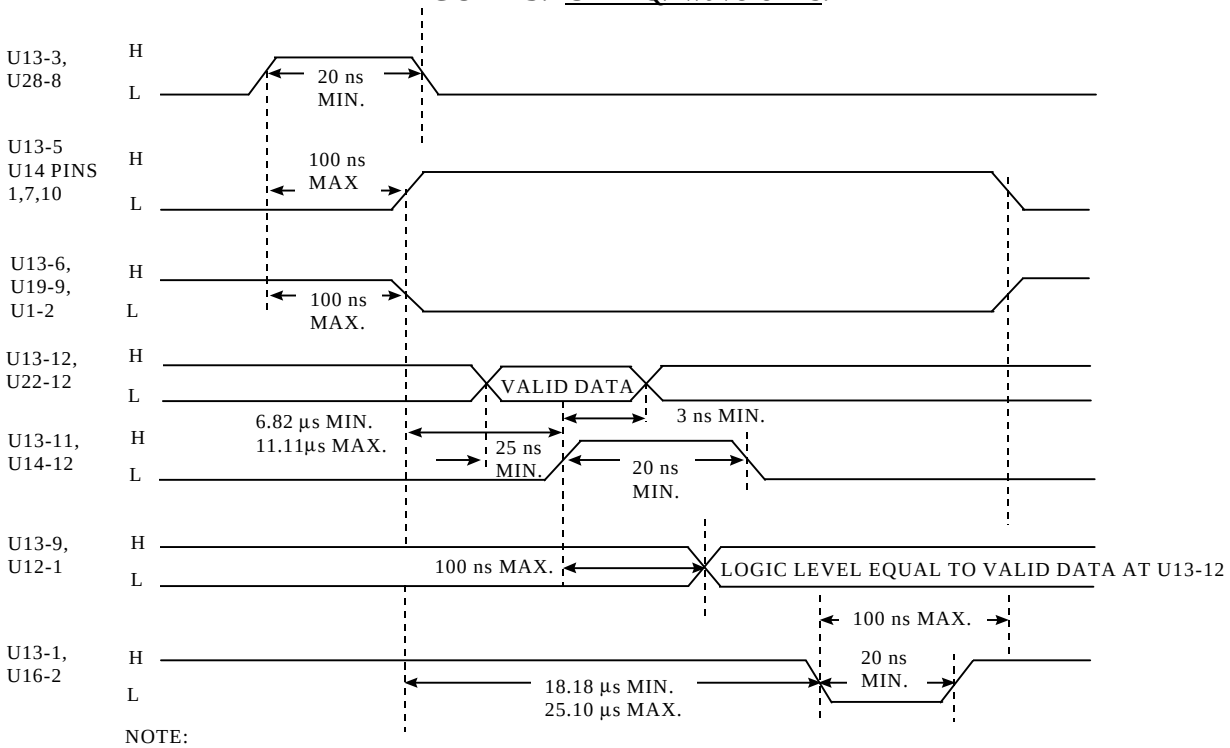
FIGURE 2. OEN-Q waveforms.

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NOTE:
H= LOGIC LEVEL 1
L= LOGIC LEVEL 0

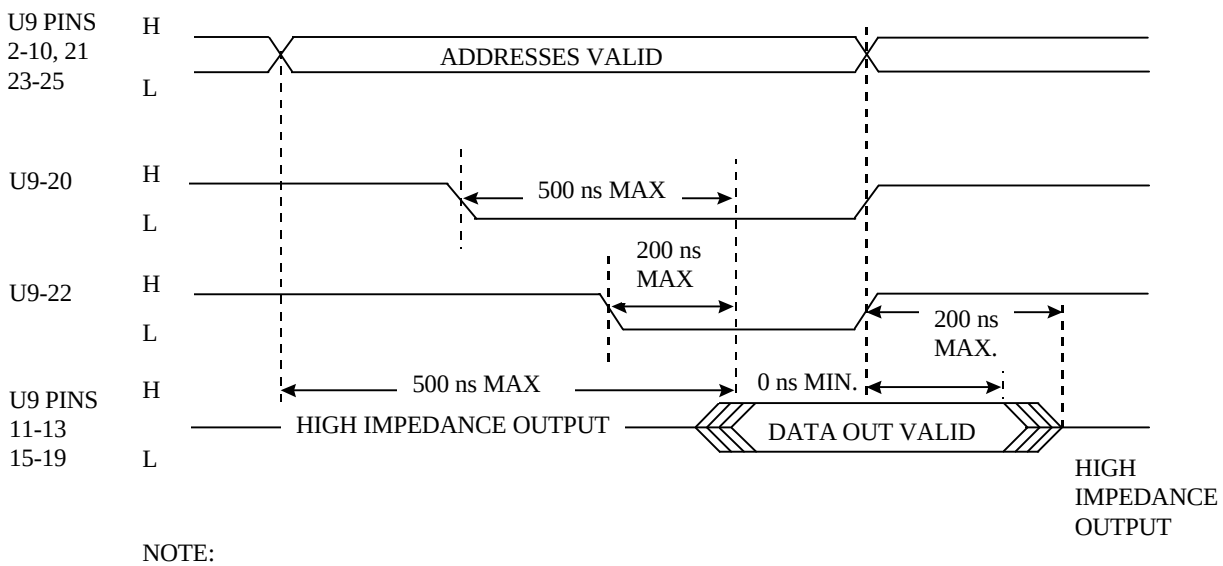
FIGURE 3. OEN-Q/ waveforms.



NOTE:
H= LOGIC LEVEL 1
L= LOGIC LEVEL 0

FIGURE 4. Microprocessor clock frequency select.

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NOTE:
H= LOGIC LEVEL 1
L= LOGIC LEVEL 0

FIGURE 5. Bootstrap EPROM read waveform.

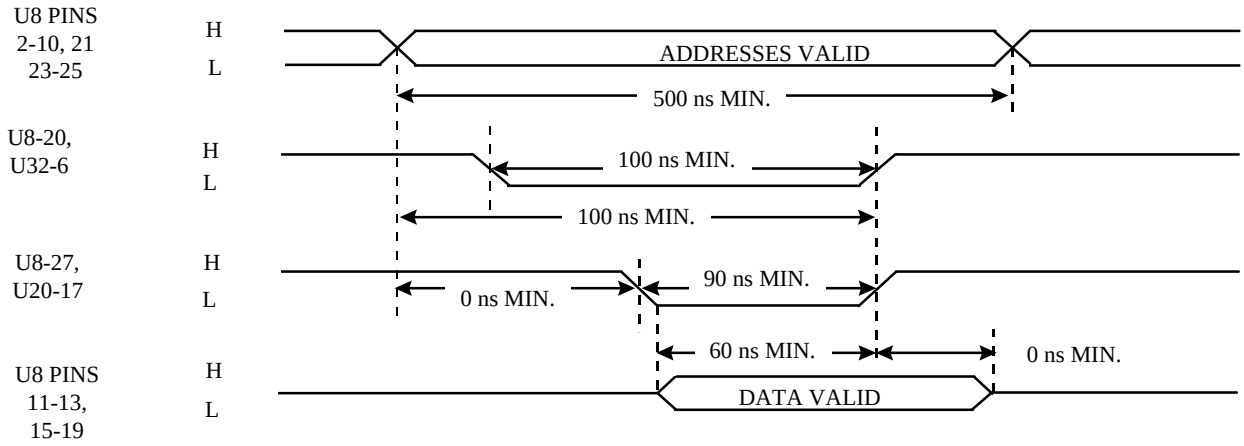


FIGURE 6. SRAM write waveforms

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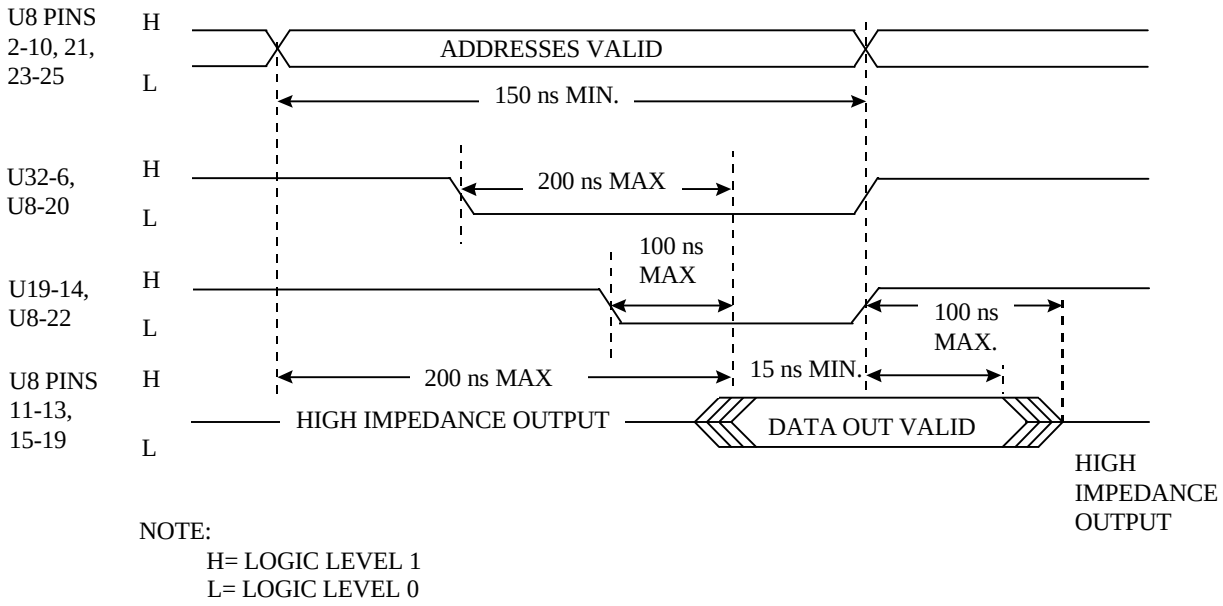


FIGURE 7. SRAM read waveforms.

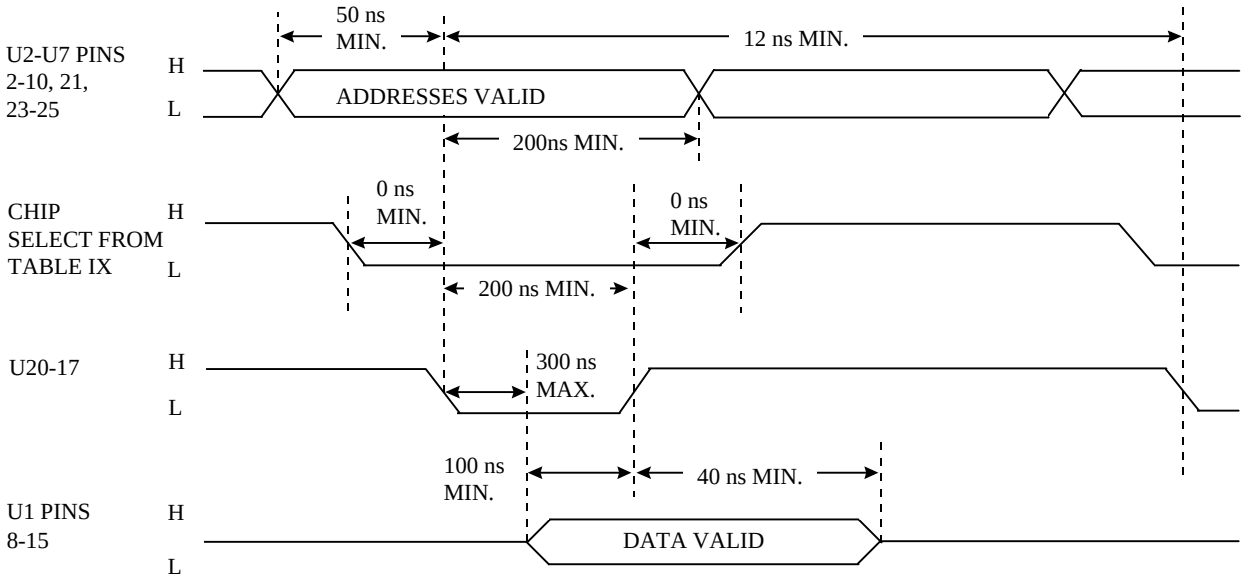
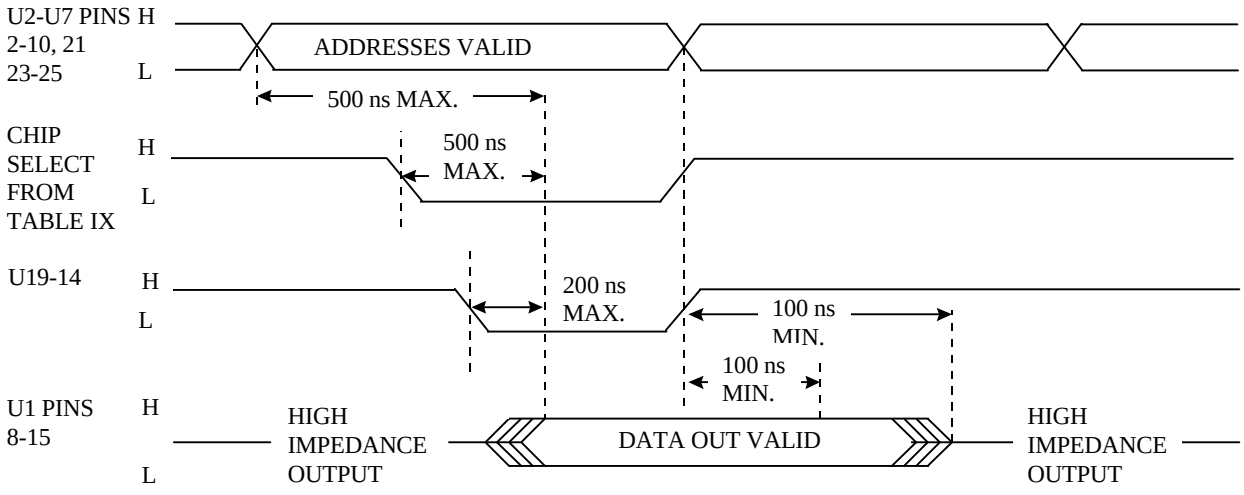


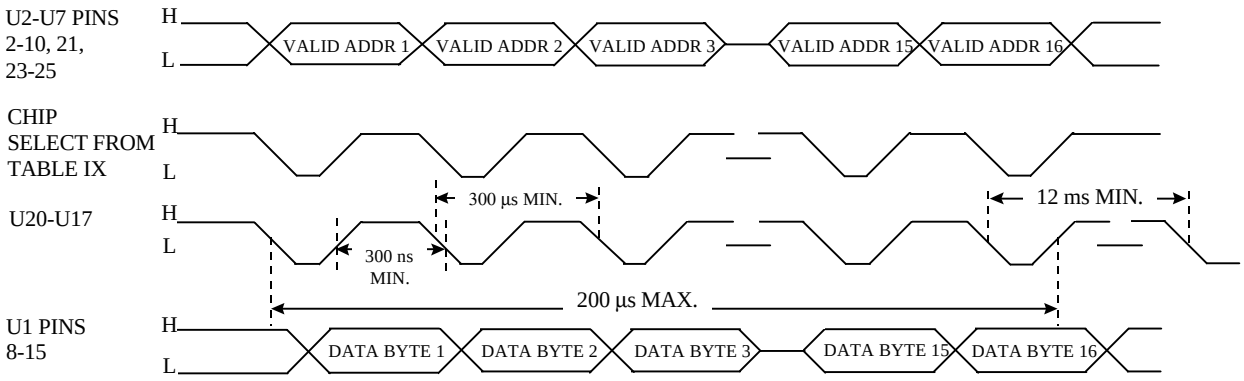
FIGURE 8. EEPROM byte write waveforms.

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NOTE:
H= LOGIC LEVEL 1
L= LOGIC LEVEL 0

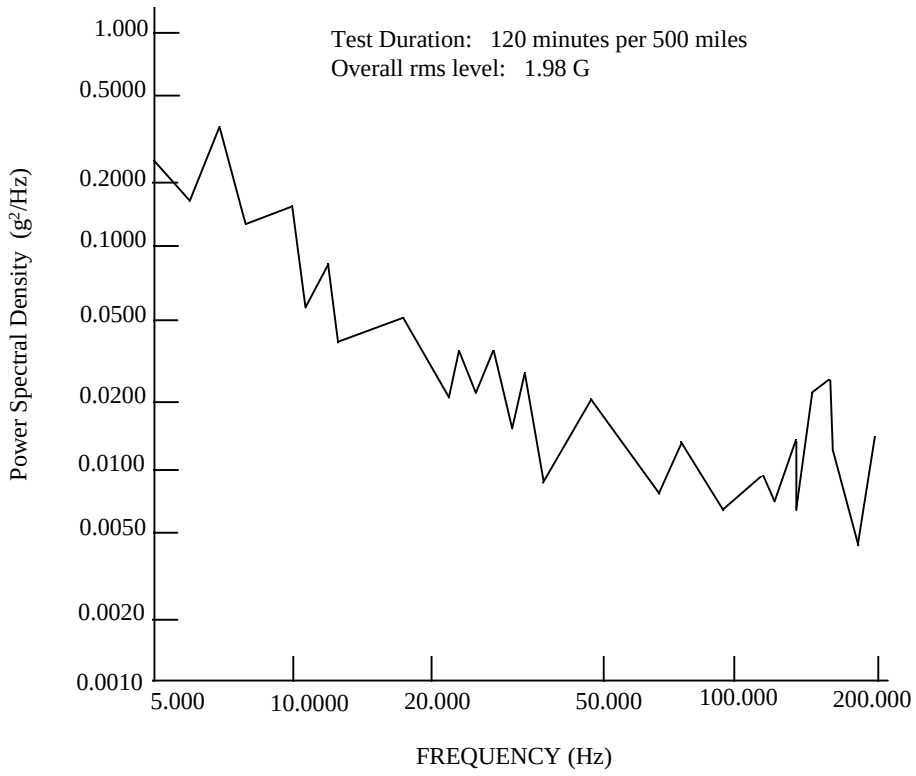
FIGURE 9. EEPROM read waveforms.



NOTE:
H= LOGIC LEVEL 1
L= LOGIC LEVEL 0

FIGURE 10. EEPROM page write waveforms.

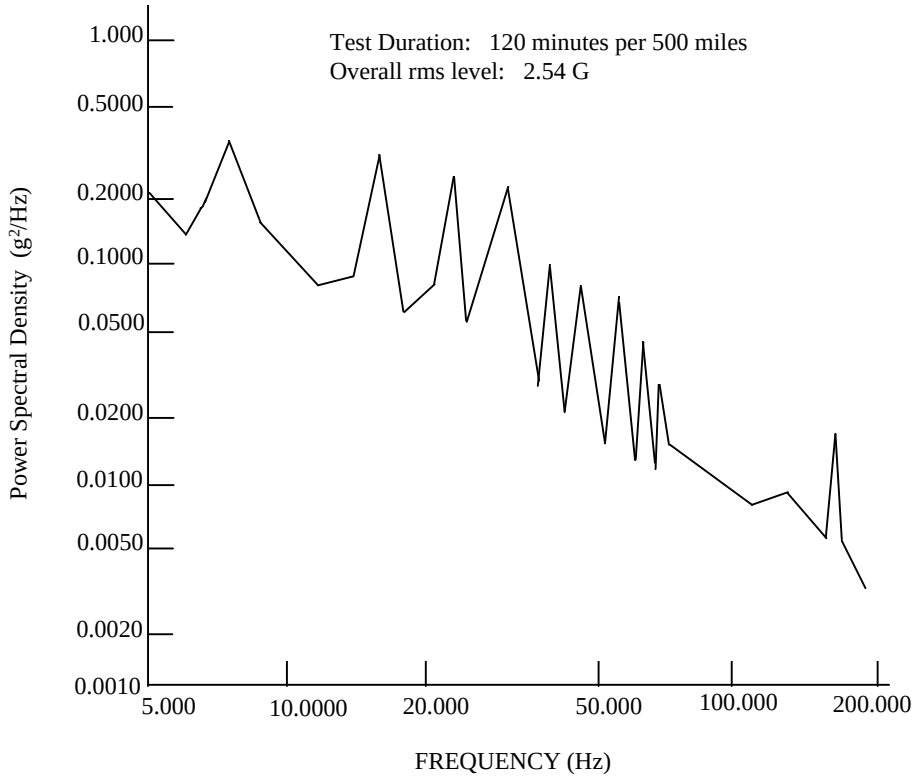
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BREAKPOINTS	
FREQ	PSD VALUE
5	0.23840
6	0.16840
7	0.38230
8	0.14870
10	0.16840
11	0.05930
12	0.09750
13	0.04090
18	0.05370
23	0.01940
24	0.03520
26	0.02310
28	0.03520
32	0.01670
33	0.03110
37	0.00810
49	0.02190
63	0.00700
73	0.01150
98	0.00660
120	0.00970
123	0.00680
138	0.01340
140	0.00760
146	0.02490
159	0.02820
163	0.01300
188	0.00480
200	0.01340

FIGURE 11. Basic transportation, composite tactical wheeled environment, vertical axis.

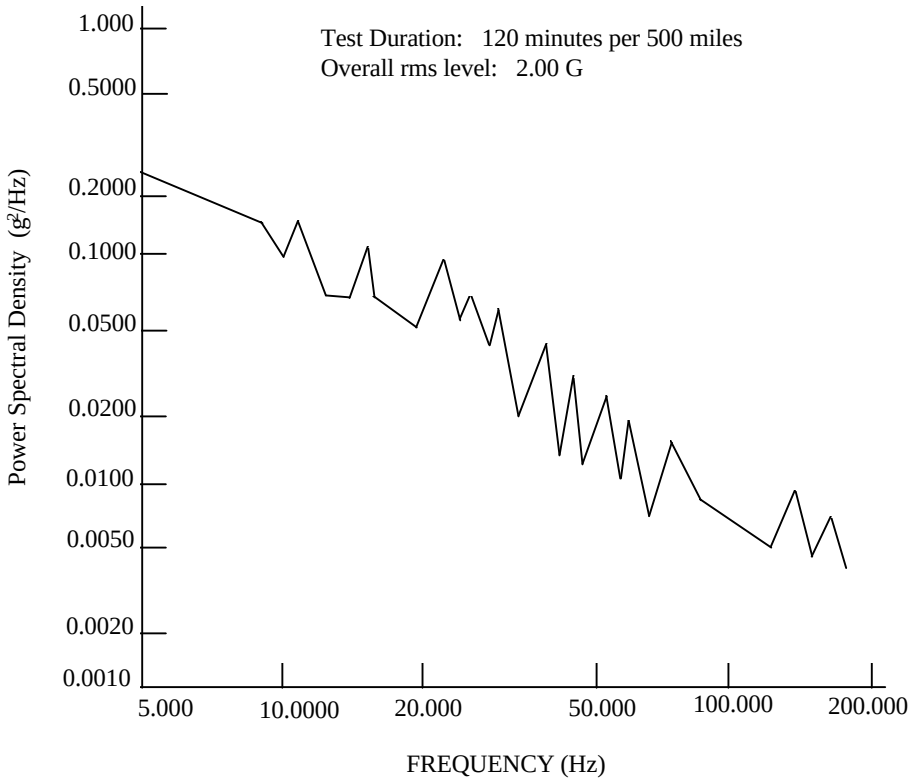
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BREAKPOINTS	
FREQ	PSD VALUE
5	0.22130
6	0.13140
7	0.20540
8	0.38230
9	0.15630
12	0.07990
15	0.08830
16	0.32940
19	0.06080
22	0.09280
24	0.26340
27	0.05640
32	0.22690
36	0.02820
40	0.11600
44	0.02090
48	0.08190
53	0.01480
56	0.07420
60	0.01270
64	0.04980
65	0.01100
68	0.03260
69	0.01480
114	0.00740
132	0.00830
160	0.00510
169	0.01710
173	0.00550
200	0.00310

FIGURE 12. Basic transportation, composite tactical wheeled environment, longitudinal axis.

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BREAKPOINTS	
FREQ	PSD VALUE
5	0.25490
10	0.13030
11	0.09670
12	0.13360
13	0.06500
15	0.06340
16	0.11230
17	0.06500
22	0.04590
24	0.08980
26	0.05070
28	0.06830
30	0.03670
32	0.05880
38	0.01920
40	0.03860
44	0.01290
48	0.02940
50	0.01140
56	0.02460
60	0.00960
63	0.01880
70	0.00660
78	0.01500
93	0.00730
137	0.00460
145	0.00910
162	0.00430
176	0.00660
200	0.00360

FIGURE 13. Basic transportation, composite tactical wheeled environment, transverse axis.

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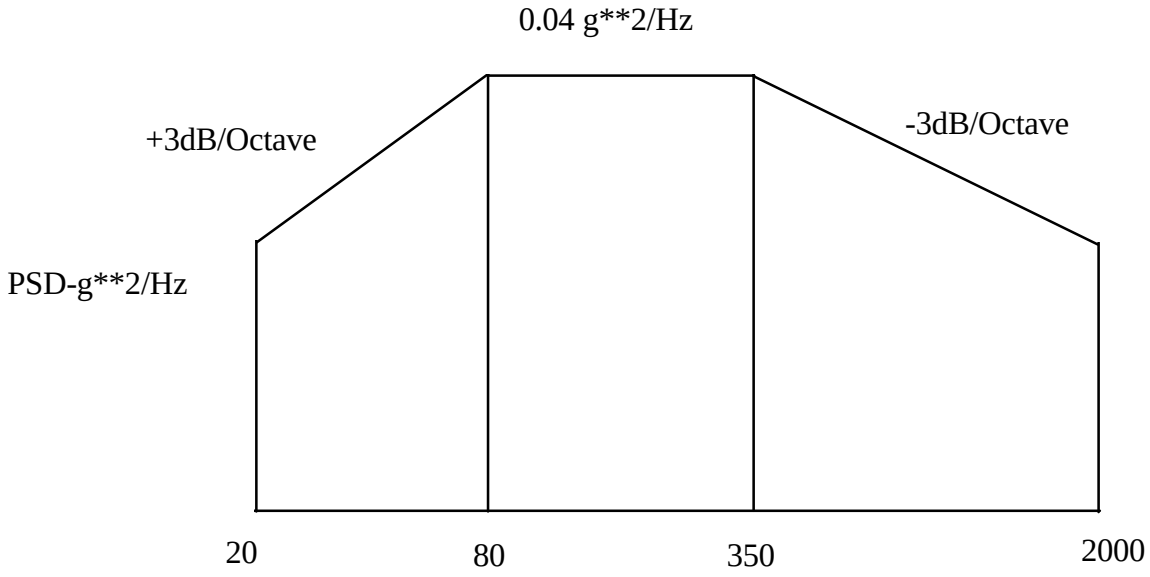


FIGURE 14. Random vibration spectrum.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 5998-0074)

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-62737A(AT)	**2. DOCUMENT DATE *(YYMMDD)*** 980123

3. DOCUMENT TITLE
Printed Wiring Assembly, Computer

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

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8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** **(810) 574-8745** **786-8745**
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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